EvoAgentBench: Benchmarking Agent Self-Evolution via Ability Transfer

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HarnessRSI 72 ★★★

evidence: retained

Paper

Data

1 What is evaluated, what is kept

Built once: Ability-supported, instance-disjoint split

1 Trace pool
3 backbones x 2 scaffolds x 2 runs, no skills, on 2,605 source tasks

2 Ability cards
Sonnet 4.6 reads traces + target: trigger, procedure, boundary, role (7,326)

3 Canonical families
3 LLM judges; experts review non-unanimous pairs, split bad groups: 170 families

4 Graph and split
a test task must share an Ability with a train task: 528 train / 267 test

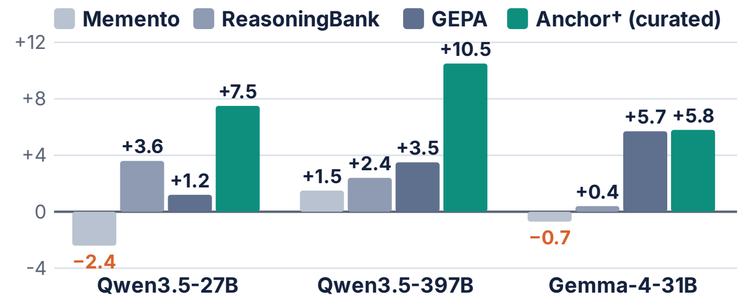
Per setting (6): evolve on train split only

- **Memento** — closest stored case
- **ReasoningBank** — distilled strategies
- **GEPA** — one evolved prompt
- **Anchor†** — curated skills, label-routed

gain = mean over test tasks of (score with evolution - Vanilla)
no accept/reject rule inside the benchmark; weights never change
3 runs per instance, mean ± SE

2 Anchor gains in 24/24 cells; 6 inside SE ‡

Average gain over Vanilla, points (Table 3)



Vanilla averages 37.5 / 49.2 / 36.7 (4 domains x 2 scaffolds). † Anchor routes by the test task's curator-side Ability label: a diagnostic, not deployable.

4 Where the gain is absent or thin

- **Memento 27B: −2.4** comes from one cell (Nanobot SWE 45.8 → 9.5, −36.3); its other 7 cells there are positive
- Gemma: GEPA **42.4** vs Anchor 42.5; GEPA beats Anchor on SWE (42.3 vs 32.7; 37.5 vs 26.8) — an automatic method tops Anchor in 3 of 24 cells ‡
- Cell SE is 2.2–5.9 points on 56–86 test tasks; no significance test; headroom picked on the construction backbones
- **Cost:** Anchor cuts turns in 4 of 6 settings (to −8.8%) but adds +7.6% and +32.7% on Gemma; GEPA adds turns in 4 of 6

3 Controls that decide if the gain is real

Per-cell view of Table 3

of 24 cells	negative gain	|gain| < own SE ‡
Memento	7 ‡	16
ReasoningBank	6	11
GEPA	4	14
Anchor†	0	6

GEPA 4 and ReasoningBank 6 negative cells are the paper's counts. The SE comparison is ours, not a test.

Matched / reported

- Same tasks, tools, timeouts, scoring, config for every method
- State from train split only; 528 / 267; access-control leakage protocol (App. B)

Not in the paper

- Never given the same teacher: Anchor is written by Sonnet 4.6 with the ground-truth target and 3 other backbones' traces; automatic methods learn from the evaluated model's own rollouts (Web 8.7–13.3 on 27B / Gemma). “The gap is routing” is inferred
- Random-split comparison; empirical leakage or near-duplicate test
- Cost / compute of evolution (only a top-10 retrieval budget is stated); whether evolution is repeated across the 3 runs

Takeaway

The benchmark's own Ability-routed skills gain in 24 of 24 cells, but 6 of those gains are below the cell SE (3 runs, no test). Memento, ReasoningBank and GEPA have negative cells. Anchor differs in teacher (Sonnet 4.6, ground-truth access), traces and routing, so the gap is not isolated to any one of them.

Our read · HarnessRSI

Score 72 · ★★★ · evidence level: retained

Reports: held-out split, 3 runs ± SE, fixed environment.
Does not report: cross-domain transfer, evolution cost, improver comparison, empirical leakage test.